Predicting Emotional Trajectories in Mental Health Forums: A Linear, Aggregate Signal in Beyond Blue Discussions

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Abstract

Public mental health forums offer an accessible and supportive space for self-help, but this environment is often unsupervised, and the quality of a response can vary heavily depending on who sees a post first, creating a risk of harm. This research asks whether the emotional trajectory of a thread can be predicted as it unfolds, both for the conversation as a whole and for the original poster's own responses. We use a private dataset from the Beyond Blue forums, spanning nine sub-forums and containing roughly 4,000 posts and 200,000 comments. Each post and comment was automatically annotated into a 43-dimensional emotion vector, and a cutoff-based prediction task was tested across several models, including ridge regression, a multilayer perceptron, an LSTM, and a transformer, while also comparing the predictive value of the emotion vectors against SBERT text embeddings. The best model reached a skill score of 0.16 above the mean-guessing baseline. While modest, this result shows that the relationship is real and, importantly, linear and aggregate: sequential modelling offered no advantage for chronological prediction, which reframes the task away from the sequence-based approach originally proposed.

1 Introduction

Access to support for poor mental health is a persistent challenge in the modern world. The stigma around seeking help is fading, and with that comes growing demand for it. Because experts such as therapists are perceived as costly, and because most people are already comfortable with social media and forums, certain online communities have emerged to help relieve this demand while providing a low-stakes entry point for self-help.

Given the vast variety of issues a person may be facing, it is difficult to provide blanket advice or responses that help the majority of people. Combinations of issues can also produce emergent negative effects that are difficult to identify unless examined by someone with both clinical expertise and detailed knowledge of the individual affected.

The crowd-sourced nature of the advice individuals receive on these platforms means the quality of a conversation can swing wildly, from genuinely supportive to potentially harmful. Since these forums often serve as a person's first attempt to reach out for help, a negative first experience can trigger a devastating retreat back into isolation. If the emotional trajectory of a conversation could be predicted before it turns, however, such discussions could be flagged for human review while there is still time to intervene, preserving the forum as a viable first step rather than a point of failure.

This paper takes these concerns and asks the question: "Can emotion detection of a conversation be used as a proper predictor for the future emotional response of a mental health forum?" Two targets are predicted: the future average emotional state of the comment section as a whole, and the future average emotional state of the original poster's own responses.

The method follows two stages. First, the text of the post and its comments is converted into an emotion vector; second, a model is trained to predict future emotion vectors. A prediction window is chosen, for example the first 50% of a thread's comments, and the average emotional response of all comments after that window is predicted. A range of configurations was tested to find the best-performing predictor, including different model architectures, input feature types, and sequential versus aggregated inputs.

The investigation shows that the predictive signal is real: a trained model can outperform baseline mean-guessing, but only when the signal is aggregated. This signal proves consistent and linear, yet it sits within a noisy, high-variance target, which is precisely why aggregation is necessary to recover it. Initial testing revealed that treating comments as a chronological sequence, rather than as directly connected turns, introduces noise that is only remedied through averaging; sequential models were therefore discarded early in the investigation.

This paper makes several contributions. It presents, to the author's knowledge, the first predictive model of emotional response on a moderated Australian mental health forum, and shows that this prediction requires only a simple linear model applied to aggregated emotion data. It further demonstrates that text embeddings do not improve predictive capacity and may actively hinder training. A per-emotion breakdown of predictability is provided, along with a comparison of predictability across the forum's different topic categories. Finally, the paper contrasts the prediction of the conversation's collective emotional response with the prediction of the original poster's individual response.

2 Literature Review

The literature reviewed here traces the development of sentiment and emotion detection and identifies the gaps this project addresses. It is organised around three themes that inform the project's design: the shift toward transformer-

based methods, the difficulty of applying them to mental health text, and the unsolved problem of domain generalisation.

2.1 The shift toward deep learning and transformers

The clearest trend across recent literature is the decline of lexicon-based and traditional machine learning approaches in favour of deep learning and transformers. This has been demonstrated at scale by applying structural topic modelling to 14,482 sentiment analysis articles from 2012 to 2024, mapping a sharp decline in lexicon methods against a steep rise in transformer-based ones, attributed to the higher labour costs of lexicon methods as datasets grow [7]. A compatible conclusion finds deep learning dominant for both health and social network data because it learns from large, unstructured text without hand-engineered features [1].

This consensus motivates the choice of a pretrained transformer as the automatic annotator here, since its attention mechanism can focus on the most emotionally salient parts of a message without the manual effort a lexicon approach demands. The field is also increasingly driven by health and social science rather than computer science alone, situating a mental-health project within a growing direction [7].

2.2 Emotion detection on mental health text

Applying these methods specifically to mental health text remains emerging and understudied [1, 2]. The obstacle raised most often is the scarcity of labelled data, identified alongside sarcasm as a primary open challenge [1]. This scarcity directly justifies the annotate-then-predict pipeline used here: because no large hand-labelled emotional dataset exists for a forum like Beyond Blue, an automatic detector generates the labels a second model is trained to predict.

Prior work confirms transformers can read the emotional content of mental health text, if imperfectly. An assessment of nine language models prompted as psychiatrists and evaluated by real professionals found safe and coherent responses but with inconsistency between models, and noted that supervised models built specifically for mental health still outperform fine-tuned general-purpose ones [5]. The relevant takeaway is that broad emotional content is recoverable, but fine-grained nuance is harder to capture reliably, a caveat that carries into the choice of annotator.

2.3 Domain adaptation and generalisation

The most persistent open problem is domain adaptation: the tendency of models to fail on data outside their training domain. One review treats this as a major and unresolved pain point, arguing that linguistic diversity, language evolution, and poor cross-domain generalisation form a lasting barrier, and points to transformer, hybrid, and self-supervised methods as the most promising responses without declaring it solved [2]. The domain shift between forums is similarly named elsewhere as a core difficulty [1].

This debate shapes a specific experimental question here. Because the dataset spans nine sub-forums, it offers a natural setting to test how well an emotional prediction model transfers across related but distinct domains. Whether a model trained on one sub-forum generalises to another, and whether pooling helps or hinders, becomes a direct, testable form of the domain adaptation question the literature leaves open.

2.4 The gap this project addresses

Together, the literature establishes that transformer-based emotion detection is mature, that mental health text is a promising but under-resourced target, and that domain generalisation remains contested. What is largely missing is prediction. The most comparable forum work compares Reddit's r/Depression with Beyond Blue and finds the latter's moderated discussions more positive and supportive, but this work is descriptive: it characterises the emotional signal rather than forecasting where a conversation's emotion is heading, and relies on older, non-transformer methods [8]. The field also skews toward Twitter and Reddit, leaving moderated and Australian forums underrepresented. The gap this project fills is twofold: it moves from describing the emotional signal in a moderated forum to predicting its trajectory, on a platform and domain existing work has overlooked.

3 Project Review

This section evaluates the prior systems most closely related to the project's task. The defining distinction is that prior systems classify a given label from text, whereas this project forecasts a future emotional state from past emotional states, a different task with no ground-truth future to train against.

3.1 LLM-based prediction from forum text

The closest analogue is the Mental-LLM work [4], which evaluates models including GPT-3.5 and GPT-4 on mental health prediction across Reddit-derived datasets. It finds prompting strategy matters little, but fine-tuning improves performance substantially, with the best fine-tuned model beating the strongest GPT-3.5 prompt by 10.9%. This confirms forum text carries predictable signal and that Reddit-style data is a conventional testing ground.

The difference is task formulation. Every dataset used is already labelled, and the models predict those labels from text: supervised classification with a defined ground truth [4]. This project has no labels and no ground-truth future; it generates emotional labels automatically, then predicts their aggregate value later in the conversation. The problem is forecasting rather than classification, which introduces the temporal cutoff design and the absence of a verifiable target.

3.2 Automatic annotation of unlabelled data

Automatic annotation of unlabelled forum text is supported by work using language models to analyse the sentiment and vaccine hesitancy of discussions from X, Reddit, and YouTube, which finds little difference between learning paradigms and recommends a hybrid approach [6]. The study demonstrates that automatically labelling unlabelled forum text is a viable foundation for downstream analysis, the strategy this project depends on.

Its limitations mark the divergence. That work focuses on vaccine discourse, whereas this project operates in the

broader and more emotionally complex mental health domain [6]. Its target of sentiment and hesitancy sits on smooth, near-analogue scales; emotion detection across many discrete categories is harder, and finer-grained targets make reliable annotation more difficult. This project tests whether the annotation strategy that worked for sentiment extends to a more demanding emotional setting.

3.3 Text mining on Beyond Blue

The most directly comparable study is the only prior work identified using Beyond Blue itself [8]. Through sentiment analysis, topic modelling, and word-category analysis, it finds Beyond Blue more positive and supportive than r/Depression, attributing this partly to its moderation. Its strength, for this project, is establishing that Beyond Blue contains meaningful, extractable emotional signals, of high quality precisely because the forum is curated. Its limitation is that the methods are descriptive and pre-transformer: the study does not model how a conversation's emotion evolves. It shows the signal is there; this project asks whether it can be predicted.

3.4 Positioning this project

A consistent pattern emerges: comparable work either predicts pre-existing labels [4], annotates text for descriptive analysis [6], or characterises content without forecasting it [8]. Two features distinguish this project. First, it is a forecasting task, predicting a future aggregate emotional state from earlier states in the same conversation rather than classifying a message in isolation. Second, where prior work commits to a single model or family, this project systematically compares model capacities, feature types, and input representations to determine not just whether prediction is possible, but what kind of model the signal requires. This combination, applied to an underexamined moderated Australian forum, defines the project's contribution.

4 Methodology

4.1 Dataset

The dataset contains roughly 4,000 posts and 200,000 comments from nine Beyond Blue sub-forums, received as 18 files (one post file and one comment file per sub-forum). Each entry included the text content, usernames, the comment-to-post association, and timestamps. Several cleaning steps were required. When combining posts into a single table, each post ID had to be reassigned, because every file began at zero and counted up, causing overlaps when merged. Timestamps were regularised, and emoji were converted to text to carry their emotional information to the parser. Usernames inside comment bodies were replaced with the neutral word "someone", preventing usernames such as "sad_flower123" from being parsed as "sad" and "flower" and distorting the signal. Rows with missing information, comments referencing non-existent posts, and posts with too few comments to be trainable were removed. A random 25% of each sub-forum was then held out as a test set, split at the thread level to prevent leakage and stratified so every sub-forum was represented.

4.2 Emotion annotation

Post and comment text was split into sentences and passed through the bert-43-multilabel-emotion-detection model, converting each sentence into a 43-dimensional vector. This model was chosen for three reasons: it is lean at only 100 million parameters, reducing parsing time; although a classifier, it outputs values between 0 and 1 across all 43 dimensions, allowing finer tracking of emotional variation and multiple simultaneous emotions; and it was trained on multiple datasets, making it more adaptable than a model trained on GoEmotions alone. The sentence vectors were aggregated by mean into a single vector per post or comment. This was not the original design. Initial testing on the sentence-by-sentence data failed: variance between sentences was so high and unpredictable that models simply predicted the average rather than learning. Averaging to the message level produces a far more stable signal and keeps the unit of analysis at a whole message.

4.3 Task formulation

The experiment predicts the average emotional vector of comments past a cutoff. For example, the model observes the post and the first 50% of comments before predicting the average vector of the remaining 50%. Three cutoffs were tested, at 25%, 50%, and 75%, to examine how accuracy changes with how early in a thread the model is applied. This aggregate framing replaced the original formulation, in which the model predicted each comment's emotion vector sequentially. That approach was abandoned after a persistence model (predicting the next vector to equal the previous one) lost to mean-guessing, most likely because the data contains large outliers that generate huge errors alongside strong central tendencies. Having shown per-comment forecasting to be near-unlearnable, the task was reformulated around the more stable average. This early testing was valuable in itself, narrowing the scope before major testing began.

4.4 Prediction targets

The model predicted two targets: the mean emotion vector of all comments not written by the original poster, and the mean vector of all comments written by the original poster, excluding the post itself. Both were taken from data after the cutoff. Posts where the poster did not respond were excluded from the second target, reducing its dataset size accordingly.

4.5 Feature representations

The primary features were the 43-dimensional emotion vector of the post and the 43-dimensional average emotion vector of the pre-cutoff comments. Testing also assessed whether adding the 768-dimensional SBERT vector (all-mpnet-base-v2, unit-normalised) improved performance. Three combinations were compared: emotion only, SBERT only, and the two concatenated. Features were standardised per block, so the emotion and SBERT vectors were normalised separately, preventing the larger vector from dominating the feature space.

4.6 Models

Five models were compared, both to find the best performer and to characterise the dataset. The baseline is a mean guesser predicting the mean vector at every sample. The others are a ridge regression with the regularisation hyperparameter swept from 0.1 to 100,000; a MLP with two hidden layers of size 256; a single-layer LSTM of size 128; and a transformer of size 128 with 4 heads and 2 layers. The LSTM and transformer were each tested in two setups: sequential, with the emotion vectors arranged in order, and non-sequential, with the vectors averaged before and after the cutoff to match the input seen by the simpler models.

4.7 Evaluation metric

Skill score measures each model's ability to beat the baseline. It is one minus the model's mean squared error divided by the baseline's, giving a range from negative infinity to 1: negative means worse than baseline, zero means equal, and 1 means no error. It was chosen because the mean guesser already achieves 0.78 accuracy, which leaves little apparent headroom and, worse, obscures failure: a model at 0.77 accuracy is actually worse than baseline, but only skill score makes this clear by producing negative values. Cosine similarity was rejected for the same reason, as every model, including the baseline, scored above 0.99; because the emotion vectors are all positive and many emotions correlate, cosine rarely falls below 0.5 and wastes most of its range.

$$\text{Skill} = 1 - \frac{\text{MSE}_{\text{model}}}{\text{MSE}_{\text{mean}}} \tag{1}$$

Two skill scores were measured. The macro score is the mean of each dimension's individual score, weighting every emotion equally like macro-F1. The pooled score uses the total mean squared error across the whole vector. Comparing them reveals whether individual dimensions disproportionately affect the result. In all results, no macro skill score deviated far from pooled skill score.

4.8 Validation protocol

The split was performed per sub-forum, and skill score was evaluated on each sub-forum individually and jointly, assessing performance both within and across domains as raised in Section 2.3. The same split and metric were applied across all conditions to keep comparisons fair.

5 Ethics

5.1 Data privacy

This information is publicly accessible, having been and continuing to be hosted on a public online forum, but the dataset contains sensitive and personal information regarding users' personal lives, mental health states, and relationships. While working on this dataset, usernames were therefore anonymised, and the uncleaned dataset will not be republished, since the textual content contains information that cannot be fully anonymised, such as names, locations, and traumatic events.

5.2 Model-error harm

This model also makes judgments about the emotional state of individuals who may be in crisis, and false predictions could lead to worse outcomes for those individuals. A model such as this should therefore never be used in an autonomous or unsupervised fashion. Any model produced from this work should, at most, flag information for human review, in order to prevent maximal harm.

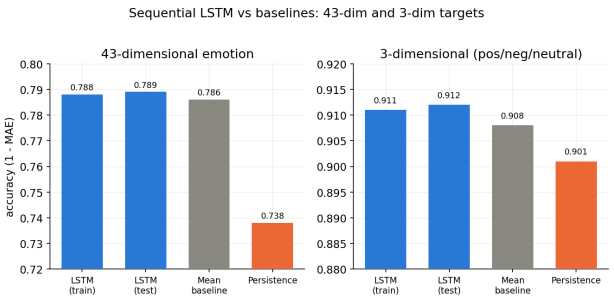


Figure 1: Initial sequential test: a standard LSTM performs at the level of the mean baseline for both the 43-dimensional and 3-dimensional targets (accuracy shown as 1−MAE; note the zoomed vertical axis).

5.3 Governance alignment

General AI ethics principles promote wellbeing, privacy, and accountability, all of which require any output from this work to be both anonymised and reviewed by accountable humans before any real-world decision is made.

5.4 A results-specific concern

Because the model predicts the emotional state of both the future conversation as a whole and the individual owner of the post, extra care must be taken over the profiling risks this creates. Unlike a system that predicts only group-level trends, this model forecasts the trajectory of identifiable individuals, which raises a sharper privacy concern that any deployment would need to address.

6 Results

6.1 Initial sequential testing

Initial testing was performed on sequential data, feeding each comment to the model one after another. A standard LSTM model performed similarly to the baseline mean model, shown in Figure 1. Reducing the feature space by binning the emotions into positive, negative, and neutral, collapsing the 43 dimensions to 3, did not change the model's relative improvement over the baseline. The persistence-to-mean MSE ratio was greater than 1.5, and at the 25% cutoff approached 1.9. On the basis of this result, the task was reformulated: rather than predicting each comment sequentially, all subsequent tests predicted the average emotion vector after the cutoff, and the average emotion and SBERT feature vectors were used as inputs, with the exception of the later sequential retests of the LSTM and Transformer.

6.2 Baselines and signal existence

Figure 2 shows the skill of the persistence model against the mean baseline across the three cutoffs. The persistence model scores below the baseline in most conditions, and its skill approaches zero as the cutoff grows. Across the three cutoffs, the mean absolute correlation between the

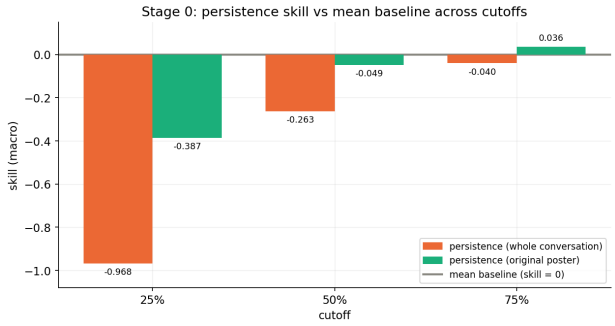


Figure 2: Skill of the persistence model against the mean baseline across the three cutoffs, for the whole-conversation and original-poster targets; the mean baseline sits at zero by definition.

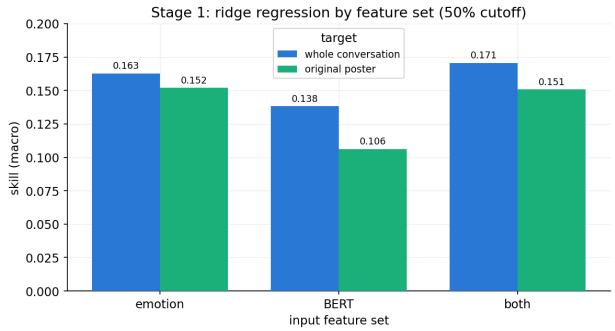


Figure 3: Ridge regression skill by input feature set (emotion, SBERT, and the two concatenated) at the 50% cutoff, for both prediction targets.

pre-cutoff context vector and the post-cutoff target vector, averaged over the 43 dimensions, ranged from 0.33 to 0.36.

6.3 Linear floor and feature ablation

Figure 3 compares a ridge regression model trained on three input sets: the average emotion vector alone, the SBERT feature vector alone, and the two concatenated. All models from this point use the skill score defined in the methodology. The SBERT vector alone produced a lower skill than the emotion vector alone. The concatenated vector improved skill by a small margin for the whole-conversation target (0.171 against 0.163) but slightly reduced it for the poster target (0.151 against 0.152). Ridge on emotion alone reached a skill of 0.163 for the whole conversation and 0.152 for the poster. Figure 4 extends this comparison to a multilayer perceptron across the same three feature sets. The MLP scored below ridge in every condition, and collapsed to negative skill on the SBERT and concatenated inputs, reaching -0.221 on SBERT for the whole-conversation target.

6.4 Per-emotion predictability

Figure 5 shows the per-dimension skill of the ridge emotion model. All 43 emotions produced a positive skill score, meaning every emotion was at least somewhat predictable above the mean baseline. The most predictable emotions were autonomy, safety, grief, curiosity, and admiration; the least predictable were surprise, worry, optimism, love, and caring. Figure 6 plots each dimension's target variance against its skill score. The regression line has a

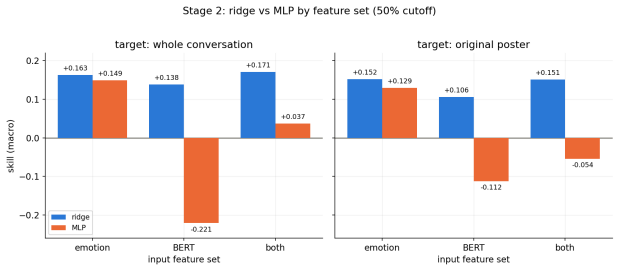


Figure 4: Ridge regression skill by input feature set (emotion, SBERT, and the two concatenated) at the 50% cutoff, for both prediction targets.

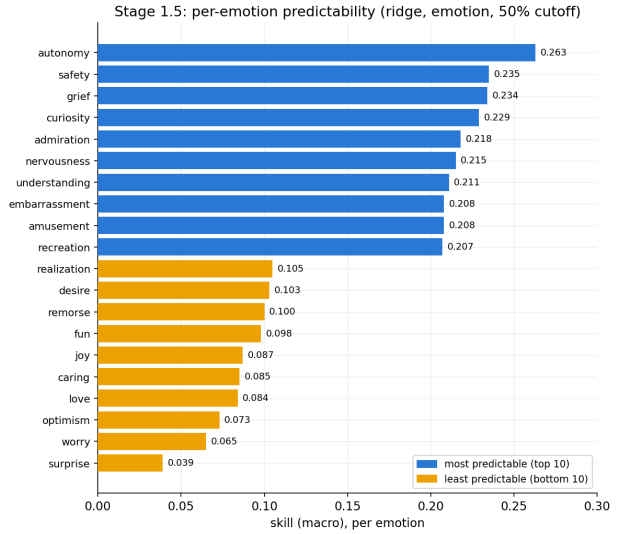


Figure 5: Per-emotion skill of the ridge emotion model at the 50% cutoff, showing the ten most and ten least predictable of the 43 emotions; all dimensions are positive.

slope of approximately 3.5 with large residuals around it. Low-variance dimensions were not systematically harder to predict above the baseline than high-variance ones.

6.5 Model comparison on averaged input

Figure 7 compares four models trained on the averaged pre-cutoff emotion vector to predict the averaged post-cutoff vector. For the LSTM and Transformer this is a single-timestep input. Skill declined with model capacity: ridge (0.163), MLP (0.148), LSTM (0.141), and Transformer (0.005) for the whole-conversation target, with a similar ordering for the poster target.

6.6 Sequence versus aggregate input

Figure 8 compares the LSTM and Transformer trained on the full pre-cutoff sequence against the ridge model on averaged input. Neither sequence model exceeded the ridge baseline: the LSTM reached 0.118 and the Transformer 0.153 for the whole-conversation target, against ridge's 0.165 on the same sample.

6.7 Cross-forum transfer

Figure 9 shows the skill of a ridge model trained on each sub-forum individually and tested against every sub-forum, alongside the pooled model. The pooled model outperformed every sub-forum specialist on its own test set, with

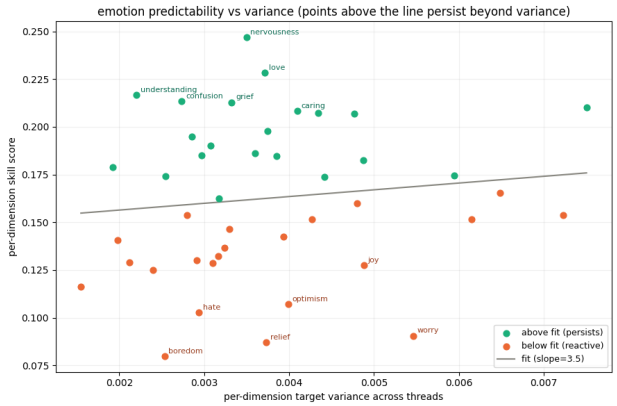


Figure 6: Per-dimension target variance against skill score, with a fitted regression line, illustrating that predictability does not simply track variance.

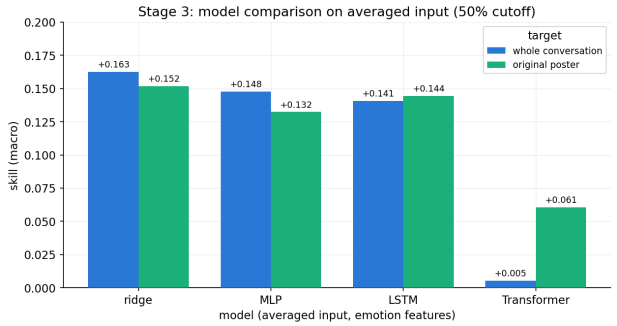


Figure 7: Skill of ridge, MLP, LSTM, and transformer models on averaged input at the 50% cutoff, for both targets; skill declines with model capacity.

the single exception of long-term-support, where the difference was 0.002. As a training source, depression produced the highest cross-forum skill; as a test target, long-term-support was the hardest for every model. The pooled row mean was 0.115, the train-on-self diagonal mean was 0.087, and the off-diagonal cross-forum mean was 0.029. Sub-forum sample size and skill were moderately related: most sub-forums below 200 training samples produced a skill improvement under 0.1, while most above 200 exceeded it.

6.8 Context-length sweep

Figure 10 shows the ridge model's skill at the three cutoffs. For the whole-conversation target, skill decreased as the cutoff grew later, from 0.184 at 25%, to 0.163 at 50%, to 0.113 at 75%. The poster target declined more gradually, from 0.159 to 0.152 to 0.128, and exceeded the whole-conversation target at the 75% cutoff. The raw mean squared error rose with the cutoff for both targets.

6.9 Contribution of post versus comments

Figure 11 shows the skill of the ridge model on three input sources: the post alone, the pre-cutoff comments alone, and both combined. For the whole-conversation target, comments alone (0.149) far exceeded the post alone (0.075), and the combined input (0.163) was only slightly above comments alone. For the poster target, the post alone (0.101) was closer to comments alone (0.120), and the combined input (0.152) clearly exceeded both.

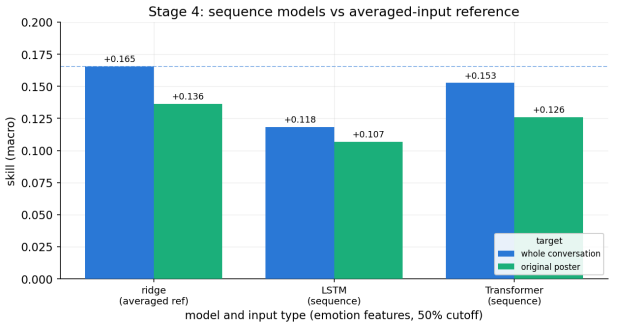


Figure 8: LSTM and transformer trained on the full pre-cutoff sequence, compared against the ridge model on averaged input (dashed line) at the 50% cutoff.

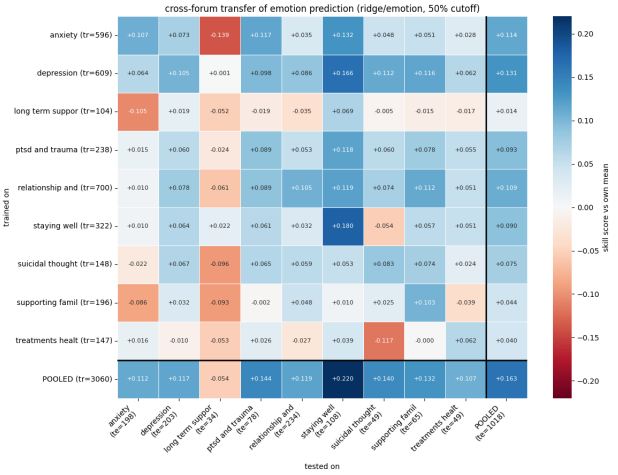


Figure 9: Cross-forum transfer matrix: rows are the sub-forum a ridge model was trained on, columns the sub-forum it was tested on, with the pooled model in the bottom row; cells show skill against each forum's own mean baseline.

7 Discussion

7.1 The central finding

The predictive signal is real but modest, and its character is the main result. Across every test, a simple linear model on the aggregate emotion vector reached a skill of roughly 0.16 over the baseline, and no more complex model improved on it. A thread's future emotional state is therefore weakly but genuinely determined by its aggregate context, not its conversational sequence. This follows from Section 6.1: with a persistence-to-mean MSE ratio greater than 1.5, comment-to-comment transitions are nearly independent, so the signal cannot live in the chronological sequence and survives only in the thread-level average, where the noise of individual comments cancels out.

7.2 Thematic versus reactive emotions

The per-emotion results suggest a consistent divide. The most predictable emotions, such as autonomy, safety, grief, and nervousness, are sustained dispositional states describing what a thread is about; the least predictable, such as surprise, worry, and joy, are momentary reactions. A thread's subject persists across its length, so the first half predicts the second, whereas reactions fire on specific comments the model never sees. Figure 5 supports this being a real

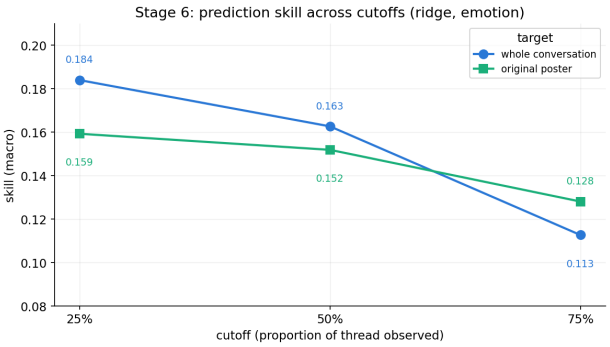


Figure 10: Ridge model skill across the three cutoffs for both targets, showing skill declining as the cutoff grows later for the whole-conversation target.

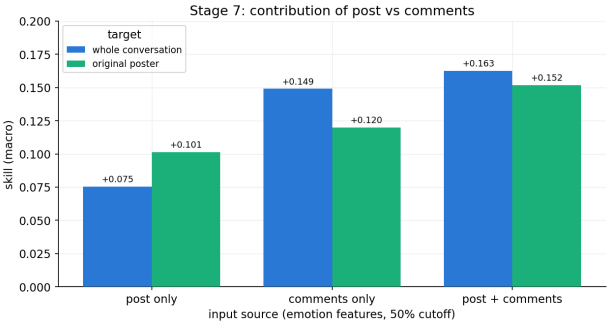


Figure 11: Ridge model skill by input source (post only, comments only, and both) at the 50% cutoff, for both targets.

effect rather than an artifact of variance. One caution is that worry and nervousness are near-synonyms yet sit at opposite ends of the ranking, which may reflect how the emotion model assigns those labels rather than a genuine difference.

7.3 The conversation versus the individual

The two targets behave differently, which speaks to the dual aim set out in the introduction. For the whole conversation the signal is emergent: the pre-cutoff comments carry roughly twice the predictive weight of the post. For the poster it is dispositional: the post alone predicts the poster's later state almost as well as the comments do, and combining the two beats either alone, a form of personal emotional consistency since post and later replies share an author. The opening post is thus a weak predictor of where the crowd goes but a strong predictor of where its author goes.

7.4 Cross-forum transfer and domain adaptation

The transfer results give a nuanced answer to the domain-adaptation problem raised in Section 2.3. A model trained on one sub-forum transfers poorly to another, with an off-diagonal skill of only 0.029, yet the pooled model outperforms every individual specialist, including on their own test sets. Both hold together because pooling recovers a shared core relationship from the added data while regularizing away each forum's idiosyncrasies. The signal that generalities is therefore a common core recoverable only at scale, rather than forum-agnostic dynamics that transfer directly, which refines rather than confirms the literature's framing of domain adaptation.

7.5 Evaluation and limitations

The key design decision to examine is the pivot from the proposed sequential formulation. The original framing was not wrong so much as aimed at the wrong level: the signal exists at the aggregate rather than the sequential level, as the early persistence test revealed, and treating this as an empirical finding rather than a failure to deliver is the honest reading. A more fundamental point concerns the annotation. This project treats the emotion encoder's output as ground truth, and every result is conditional on that. This was deliberate: no large hand-labeled emotional dataset exists for a forum of this kind, the labeled-data scarcity the literature identifies, so no human ground truth was available; and treating the encoder as correct keeps the pipeline tractable, letting the work focus on prediction rather than validating an emotion model. The cost is that any systematic bias in the encoder propagates into every result, as the worry-versus-nervousness anomaly hints. The predictions are best understood as predictions of what the encoder reports, a reasonable proxy for emotional state but not a verified measurement. The skill ceiling is also modest at around 0.16, the sequence models were not exhaustively tuned so a smaller gap to ridge is possible, and the smallest sub-forums carry wide uncertainty, particularly long-term-support at 34 test threads.

7.6 Recommendations for stakeholders

For a platform such as Beyond Blue, these results support a specific and limited use. Aggregate emotion prediction is a viable signal for monitoring emotional trends at the community level, but the ceiling of around 0.16 means it cannot reliably flag any single conversation. The appropriate deployment is a cohort-monitoring aid that surfaces trends for human review, not an autonomous per-thread flagging system, consistent with the human-in-the-loop position of the ethics statement and the harm the introduction set out to address. The poster's separately predictable trajectory is promising for identifying at-risk individuals, but it sharpens the earlier profiling concern, and any such use would require explicit consent and human oversight.

8 Conclusion

This project set out to answer whether emotion detection can serve as a proper predictor for the future emotional response of a mental health forum. The answer is yes, but weakly and not in the way originally expected. Encoded emotions from mental health forums can be predicted above a naive baseline, and these predictions are best produced not by the sophisticated sequential models first proposed, but by a simple linear model, which reached a maximum skill score of 0.16 over the mean-guessing baseline.

The path to that answer was itself informative. Early testing showed that this kind of information is best predicted in aggregate rather than comment by comment, which prompted a pivot away from sequential prediction. From there, a linear model proved the best fit for the task, outperforming every more complex alternative. Along the way, the project established that aggregate feature representa-

tions beat sequential ones, that earlier cutoffs yield better predictions than later ones, that including both the post and the comments improves prediction, and that BERT-style text embeddings do not measurably help.

These findings matter because they characterise the predictability of forum emotion in a specific and previously unmapped way. Every one of the 43 emotions showed some learnable component regardless of its variance, and the predictable emotions were consistently the sustained, thematic ones rather than momentary reactions. The contribution is therefore not only that prediction is possible, but a clear account of exactly what kind of prediction the data supports.

The practical value follows directly. A model of this kind, built on a larger dataset, could underpin an automated flagging system that human moderators use to identify at-risk threads and investigate them before lasting harm occurs, all from a simple linear model kept firmly under human oversight. Future work could collect datasets that preserve the tree structure of forum threads rather than flattening them chronologically, since not every commenter has read every prior comment. Such data could support a model that predicts the likely response at each leaf of the conversation rather than an aggregate over the whole thread, potentially resolving the near-independence between chronologically adjacent comments by instead predicting genuine reply-to-reply responses.

9 Word Count

4852 Words

10 AI Usage Declaration

Claude Opus was used in the planning stage of the report to give recommendations on structure and on what was missing in initial drafts. Additionally Claude was used to condense certain sections of the report to increase clarity, improve overall readability, and reduce unnecessary overexplaination. While the data was collected manually, Claude was used to generate the code used to graph the data in order to improve the visual quality of the graphs. All values in the graphs were double-checked. All corrections to grammar or rewording recommendations were also checked. The final work is written, checked, and owned by the author.

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